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ORIGINAL INVESTIGATION

Immediate Effects of Overnight Long-Haul Travel on Physical and Cognitive Performance and Sleep in Professional Male Rugby Union Players (a 2-Part Study)

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Background: Travel can disrupt sleep and reduce perceived recovery in athletes. While the impact of travel on physical performance has been studied, its effect on cognitive performance remains unexplored. ***Purpose:*** This study investigated the consequences of overnight long-haul travel on physical (part I) and cognitive performance (part II) in professional rugby players traveling overseas for competition. ***Methods:*** In part I (n = 24), vertical-jump performance was assessed across 5 mornings surrounding travel (HOME, TRAVEL, and AWAY 1, 2, and 3). In part II (n = 24), mental fatigue and psychomotor vigilance (defined as the ability to respond to stimuli effectively) were assessed across 3 days (HOME, TRAVEL, and AWAY 1). Sleep was monitored on all nights preceding performance assessment using activity monitoring. ***Results:*** Compared with HOME, TRAVEL reduced sleep quantity (part I. −219 [112] min; part II, −334 [233] min; both $P < .001$) and elevated in-flight mental fatigue (part II, +12 [18]; $P < .05$). Psychomotor vigilance was impaired during TRAVEL (+8% [6%] in valid response latencies ≤500 milliseconds; $P < .001$), and reactive jump strength was reduced at AWAY 1 (−7% [10%] in drop jump; $P < .001$). While psychomotor vigilance was restored by AWAY 1 (after hotel sleep), reactive jump strength remained diminished until AWAY 3 (−12% [9%] in drop jump compared with HOME; $P < .001$) after travel. ***Conclusion:*** For teams traveling on long-haul journeys overnight for competition, it is recommended to arrive more than 3 days before matches to enable sufficient physical and cognitive recovery at the new destination to optimize performance away from home.

Keywords: vertical-jump ability, psychomotor vigilance, mental fatigue

Professional rugby players frequently endure long-haul air travel (herein, travel) and congested fixture schedules with minimal recovery time.[1] Traveling players are susceptible to jetlag, a circadian-rhythm desynchronization due to time-zone shifts, and/or travel fatigue from stressors such as inadequate sleep, exposure to mild hypoxia, cabin discomfort (ie, seating/space), and the inconvenience of airport procedures.[2]

Research on traveling athletes primarily focuses on the combined impact of jetlag and travel fatigue commonly causing sleep disruptions that impede perceptual recovery.[2] The magnitude of travel-induced sleep restrictions remains debated, contingent on travel direction, duration, and timing.[2,3] However, it is suggested that long-haul and/or overnight travel is more likely to exacerbate sleep disruptions, resulting in self-reported jetlag and fatigue.[3]

Few studies have explored the consequences of overnight, long-haul travel on physical performance.[3–6] Travel spanning 21 to 30 hours across multiple time zones, involving an 8 to 16 hours circadian shift, can deteriorate aerobic (Yo-Yo Intermittent Recovery Test) and anaerobic capacities (maximal jump and sprint performance) in elite athletes and nonathlete cohorts for 2 to 3 days after arrival.[3–6] This is particularly relevant for professional rugby players, who often compete within 2 to 3 days of arrival,[1] raising concerns about the negative effects of long-haul travel on performance.

In rugby union, the common game movements (eg, scrums, rucks, mauls, and lineouts) rely on explosive lower limb efforts requiring an efficient stretch-shortening cycle (SSC). Execution of fast SSC movements depends on greater muscular stiffness and elasticity, a quick stretch-reflex response, and greater muscular activation.[7] Greater SSC efficiency can enhance force production rate and capacity during these explosive movements that are crucial in the game.[8] Accordingly, quantifying the SSC through vertical jump assessments reflects a player's neuromuscular status while offering practitioners a way to quantify physical performance.[9]

While researchers continue to explore the effects of travel to optimize physical performance for matchday, the impact on mental fatigue and cognitive performance should not be overlooked.[10] Mental fatigue, often associated with travel demands,[11,12] can be further exacerbated by sleep deprivation,[13] which remains unexplored.

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The game demands of sports like rugby union rely on neurocognitive and psychomotor efficiency.[14] Psychomotor vigilance is a cognitive component that is consistently and significantly affected following full-night sleep deprivation, notably in nonathletic cohorts.[13] While studies of partial sleep restriction on psychomotor vigilance yield equivocal results,[15] it appears that the 3 constructs—psychomotor performance, sleep, and susceptibility to mental fatigue—may be interrelated.[10,11,13] Given that overnight travel significantly disrupts sleep,[16,17] an investigation into the consequences of mental fatigue and cognitive function may provide valuable insights for coaches and practitioners to maximize player readiness when competing away from home.

Therefore, the aim of this study was to investigate the impact of overnight, long-haul travel on sleep and physical and cognitive performance in professional rugby players. We hypothesized that travel would disrupt in-flight sleep and thereby decrease performance. However, we anticipated a compensatory sleep rebound upon arrival, with a corresponding return to baseline in both physical (part I) and cognitive performances (part II).

Methods

Participants

Forty-three professional male rugby union players (age = 23.6 [3.0] y; height = 186.6 [7.1] cm; weight = 103.2 [12.6] kg), contracted to a Super Rugby team, volunteered to participate in either part I (n = 24) and/or part II (n = 24) of this investigation, consisting of 2 separate travel journeys for competition. Before travel, all participants completed a 4-week training block, involving 4 field (rugby; 6.5 h) and 7 gym sessions (7.5 h) weekly. Five players participated in both travel journeys (part I and II). The study obtained informed consent and was approved by the University of Western Australia Human Research Ethics Office (part I: 2023/ET000065; part II: 2023/ET000851).

Overview of Study

This study comprised 2 parts, investigating the effects of overnight, long-haul travel on physical (part I) and cognitive performance (part II) using a prospective observational study design. The purpose of travel for both parts was to compete in matches. While participants were from the same team, practical constraints resulted in a time gap between assessments (part I: November 2022 and part II: August 2023), leading to the collation of 2 distinct samples.

In part I, physical performance was assessed through vertical jump assessments across 5 mornings at HOME (2 d pretravel) and AWAY (3 d posttravel; Figure 1). In part II, cognitive performance and mental fatigue were assessed across 3 days: at HOME (pretravel), TRAVEL (last hour in-flight), and AWAY (posttravel; Figure 1). Sleep was assessed using wrist-based activity monitors on nights preceding performance assessment (part I: 6 nights; part II: 3 nights), except for the night before HOME cognitive assessment in part II.

One week before each travel journey, participants were familiarized with activity monitors, subjective assessment tools, and performance tests. Baseline values (HOME) were compared with values obtained during travel (TRAVEL) and upon arrival (AWAY 1, 2, and 3; Figure 1). For part I, HOME values were established by calculating the average jump scores from the 2 consecutive mornings preceding travel and sleep scores from the 3 consecutive nights preceding travel. All testing sessions, including familiarization, were administered and supervised by the same researcher.

Travel

Travel in both parts I and II were both categorized as overnight, long-haul travel. Both journeys departed from Perth, Australia (GMT + 8) on an initial ~4 hours flight followed by a ~3 to 4 hours layover in Sydney, Australia (GMT + 11). Subsequently, players proceeded to their respective destinations (part I: Tokyo, Japan [~10 h flight; GMT + 9]; part II: Johannesburg, South Africa [~14 h; GMT + 2]; Figure 1). Therefore, travel in part I primarily induced travel fatigue due to minimal circadian shift (+1 TZ), whereas travel in part II induced both travel fatigue and jetlag (−6 TZ). All travel arrangements were in economy class. No travel management strategies were implemented in part I, but all players received a travel pack (including compression socks [worn by everyone] and hydration supplements [taken ad libitum]) in part II.

Physical Performance: Vertical-Jump Assessment

Physical performance (part I) comprised 2 tests: the drop jump (DJ) and the multirebound jump (MRJ). Participants completed a standardized warm-up[18] of 10 bodyweight squats and lunges per leg, followed by 3 submaximal countermovement jumps. Subsequently, participants completed 3 maximal DJs from a 30-cm hardtop box, with their hands on their hips, followed by an MRJ involving 10 consecutive bilateral hops. Participants were cued to drop and jump *as high* and *as fast* as possible off the ground for the DJ, to keep their knees *as stiff* as possible, and to have *as brief a contact time* for the MRJ.[19]

Ground contact times were recorded using a dual force plate system (ForceDecks FDLite, version 2.0, VALD) sampling at 1000 Hz into proprietary software (version 2.0.8245). This has been established as a reliable tool for measuring kinematic and kinetic variables in vertical jump tests.[9] The DJ with the highest reactive strength indices (RSI) was recorded for statistical analysis. For the MRJ, the 5 rebounds with the longest flight time were extracted to calculate leg stiffness.[20] Testing occurred between 07:30 AM and 08:30 AM local time. Players were not assessed on the day of TRAVEL, as players did not arrive at the testing location until 12:00.

Cognitive Performance: Psychomotor Vigilance

Cognitive performance (part II) was evaluated via a validated[21] 3-minute computerized psychomotor vigilance task (PVT; Inquisit 6, Millisecond). The software was downloaded on each participant's smartphone, which instructed them to respond to stimuli that appeared randomly with an interstimulus interval varying between 2 and 10 seconds *as quickly and accurately as possible.* Each test commenced with a habituation phase, which was omitted from analysis.[15] Outcome measures reported in this study are mean ≤500 milliseconds reaction time (RT), as well as fastest and slowest RT (categorized by the 10th and 90th percentile) for all valid response latencies. Further, errors in responses were represented by false starts (number and RT) and lapses (number, mean, and cumulative RTs). Players performed this test in silence, under the supervision of research staff. Cognitive performance was assessed between 02:00 PM and 03:00 PM local time for HOME (destination: 07:00 AM to 08:00 AM) and TRAVEL (origin: 08:00 PM to 09:00 PM). Due to training arrangements on AWAY 1, assessments occurred between 07:30 AM and 08:30 AM local time (origin: 01:30 PM to 02:30 PM).

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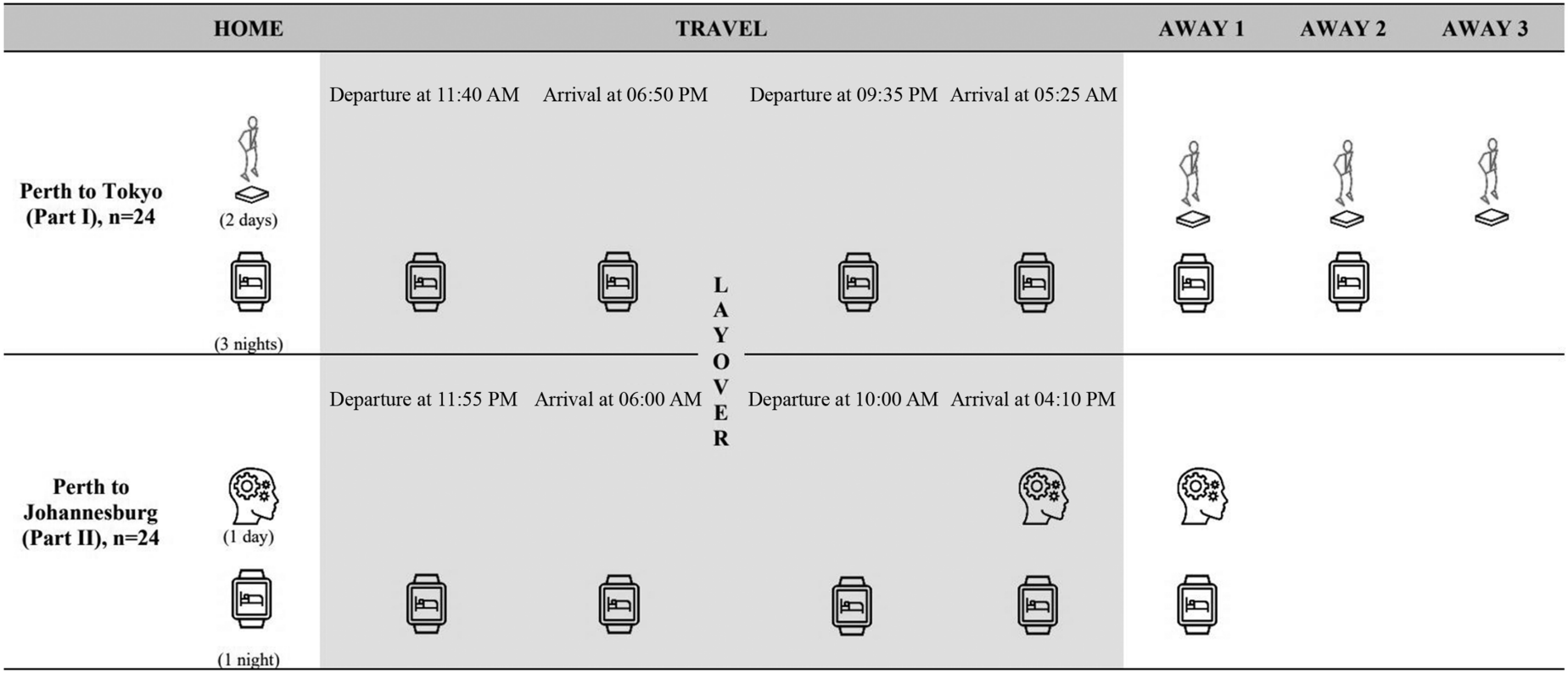

Figure 1 — Study design, including travel schedule (expressed in local time) taken by professional rugby union players to the 2 match destinations (parts I and II): Assessment of physical performance (vertical-jump ability), cognitive performance (psychomotor vigilance), subjective fatigue (visual analog scales), and sleep (wrist activity monitoring). TZ presents the time-zone difference between the departure and arrival destination. All times are presented in local time of the location.

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Perceptual Measures: Mental and Physical Fatigue and Readiness to Train

In part II, participants rated their mental and physical fatigue using 100-mm visual analog scales (VAS) anchored from "none at all" (0) to "maximal" (100).[12] Similarly, "readiness to perform" was collected, with anchors "not at all ready" (0) to "maximally ready" (100).[12] Participants completed each VAS privately, in silence, to mitigate the potential for social-influence response bias.[12] Sport-specific examples were provided to contextualize anchors for each VAS.[12] All VAS were completed immediately before the PVT assessment.

Sleep

In both parts I and II, participants wore a wrist-based activity monitor (GTX9, ActiGraph) during the nights of data collection. Sleep diaries were completed each morning in part I, with a modified version completed in-flight for part II.[22] Diaries were used alongside raw actigraphy data (60-s epoch AGD files) to manually score sleep (ActiLife, version 6.13.4, ActiGraph), ensuring optimal validity of objective sleep.[23] Sleep quantity (total minutes spent asleep), quality (surrogate marker; sleep efficiency [percentage of total time spent asleep divided by the total time in bed attempting to sleep]), and wake after sleep onset (total minutes spent awake after sleep onset) were reported. Sleep was collected across 6 nights for part I (3 baseline nights averaged for HOME, during TRAVEL, and the 2 nights posttravel [AWAY 1 and 2]) and 3 nights for part II (HOME, during TRAVEL, and 1 night posttravel [AWAY 1]).

Statistical Analysis

An a priori power analysis using previous literature[5] ($\alpha = .05$, statistical power = 90%, and effect size [*d*; derived from vertical jump performance] = 0.90) determined a minimum sample of 14 participants.

Data is presented as mean and SD. Statistical analyses were conducted separately for each data set (parts I and II) using RStudio (Version 4.0). Linear mixed-effect regressions (lmer) were used to analyze the effect of travel on performance and sleep across a fixed covariate effect of "time-point." To account for the often-reported interindividual variability among team sports, the model accounted for individual differences (*player* as a random intercept) within the team, using the linear mixed-effects models "Eigen" and S4 (lme4) package. Statistical significance was set at $P < .05$. Post hoc pairwise comparisons were performed with *Tukey*-adjusted *P* values using the estimated marginal means (emmeans) package. Effect sizes were calculated and reported as partial eta squared (η_p^2; *small* = .01, *moderate* = .06; *large* > .14).[24]

Results

The study collected 79% of jump, 69% of PVT, and 86% of perceptual scores out of the planned data points. For sleep assessment, data were gathered for 164 out of 216 nights. Missing data were due to injury (exemption from jump assessment), illness (nonattendance), and nonadherence to wearing sleep monitors.

Physical Performance

Scores from DJ and MRJ are presented in Figure 2 and Table 1. Compared with HOME, DJ RSI reduced on AWAY 1 (–8% [10]%; $P = .002$) and 3 (–12% [9]%; $P < .001$), while DJ flight time remained lower on all 3 mornings (AWAY 1: –5% [6%], $P = .014$; AWAY 2: –5% [7%], $P = .014$; AWAY 3: –6% [8%], $P = .003$). While MRJ leg stiffness remained unchanged across days ($P = .431$), MRJ flight time significantly reduced on AWAY 1 compared with HOME and all other nights (HOME: –44% [14%]; AWAY 2: –45% [17%]; AWAY 3: –33% [31%]; all $P < .001$).

Cognitive Performance

Scores derived from PVT assessment are presented in Figure 3 and Table 1. During TRAVEL, valid response latencies were significantly slower compared with HOME (mean RT ≤ 500 ms: +8% [6%], $P < .001$; 90th percentile RT: +7% [9%], $P = .001$). False starts (number: $P = .945$; RT: $P = .235$) and lapses (number: $P = .235$; mean and cumulative RT: $P = .891$ and 0.235, respectively) remained unchanged.

Mental Fatigue

Mental fatigue, as well as physical fatigue and readiness to perform, are presented in Figure 3 and Table 1. Mental fatigue was highest during TRAVEL (65 [13]), significantly increasing from HOME (+12 [18], $P = .017$), while physical fatigue did not change. Both mental and physical fatigue reduced on AWAY 1 compared with HOME (–20 [25] and –25 [21], respectively) and TRAVEL (–32 [20] and –21 [19], respectively [all $P < .001$]). Readiness to perform was significantly reduced during TRAVEL compared with HOME (–28 [22], $P < .001$) and AWAY 1 (–41 [21], $P < .001$).

Sleep

Sleep quantity and quality from both parts I and II are presented in Table 2. In both parts, sleep quantity reduced during TRAVEL compared with HOME (part I: –219 [112] min, part II: –334 [233] min), AWAY 1 (part I: –287 [96] min, part II: –256 [82] min), and AWAY 2 (part I: –263 [78] min; all $P < .001$). Similarly, sleep efficiency reduced during TRAVEL compared with HOME (–19% [15%]), AWAY 1 (–23% [14%]), and AWAY 2 (–20% [12%]; all $P < .001$) in part I only but remained unaffected in part II ($P = .268$). Significant differences in wake after sleep onset were observed on AWAY 1 compared with TRAVEL in both parts I (–37 [73] min, $P = .031$) and II (+27 [60] min, $P = .030$).

Discussion

Our findings confirm previous reports,[16,17] as overnight, long-haul journeys (parts I and II) induced partial sleep deprivation during TRAVEL (~3 h). This outcome may help explain the immediate deterioration in reactive jump strength (part I) and impaired psychomotor vigilance (part II). Although cognitive function (psychomotor vigilance) appears to be restored following a night of sleep at the arrival destination (part II), reactive jump strength remained diminished for 3 days (part I). This raises concerns for professional rugby players traveling long distances if arriving 2 to 3 days before matches. Despite a compensatory rebound in sleep quantity upon arrival (~7.5 h; parts I and II), residual fatigue persists (part II), potentially giving insight into the diminished scores of reactive jump strength (part I).

Physical Performance

Reduced DJ RSI suggests that overnight, long-haul travel undertaken in part I likely deteriorates fast SSC abilities for at least

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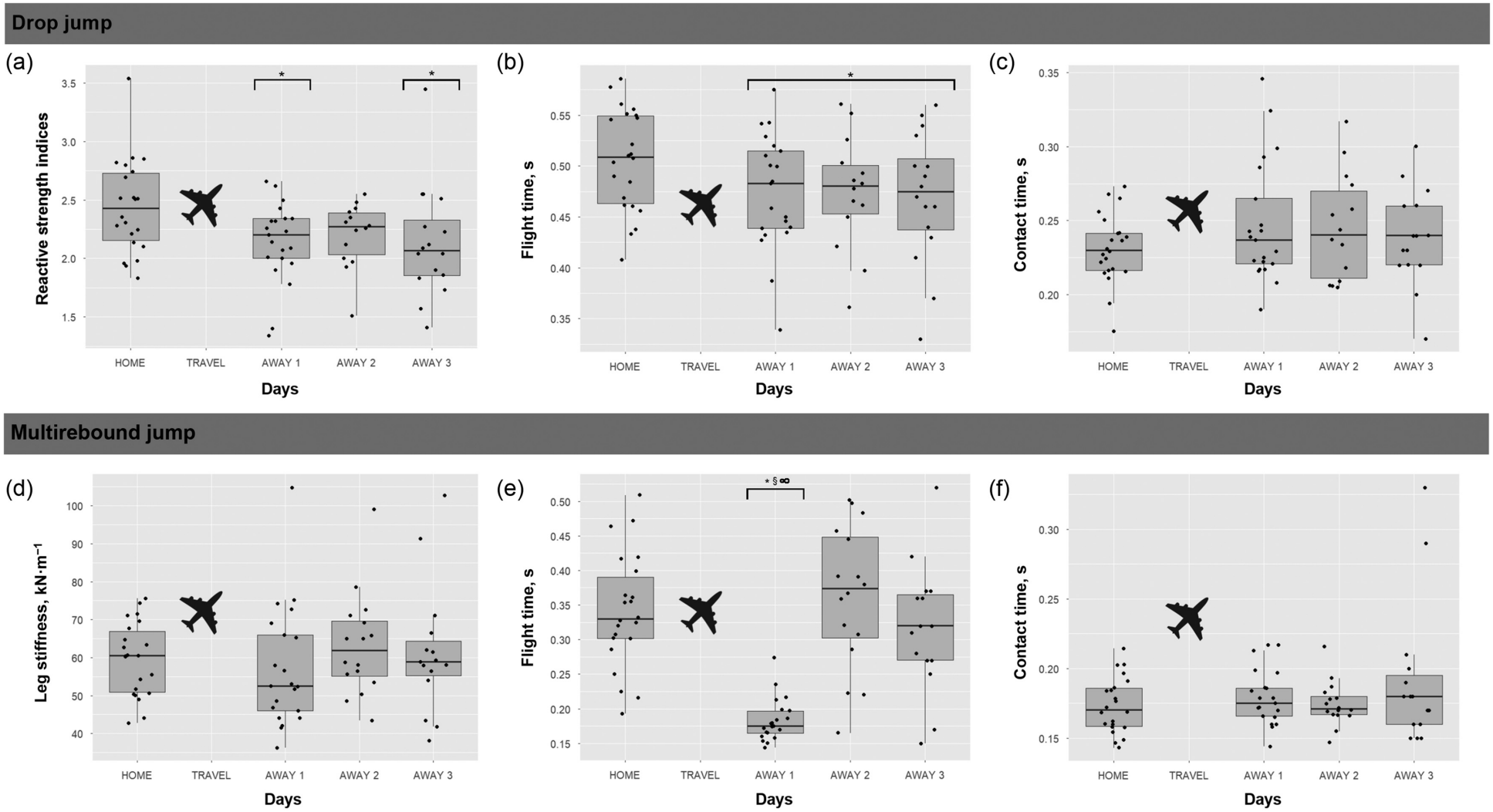

Figure 2 — Drop-jump (a–c) and multirebound-jump (d–f) variables before (HOME) and after (AWAY 1, 2, and 3) travel (part I). Outer margins of the box plots represent the interquartile range of the data, the bold horizontal lines represent the medians, the whiskers represent the minimum and maximum range 1.5 times the interquartile range of the data, and the dots represent individual data points. *Significant difference compared with HOME. ∞Significant difference compared with AWAY 2. §Significant difference compared with AWAY 3.

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Table 1 Physical and Cognitive Performance and Perceptual Measures of Professional Rugby Players Before, During, and After Overnight Long-Haul Travel to 2 Match Destinations (Parts I and II)

	HOME	TRAVEL	AWAY 1	AWAY 2	AWAY 3	ANOVA *P* value	η_p^2	Effect size
Physical performance (part I)								
DJ RSI	2.44 (0.40)		2.14 (0.34)[a]	2.20 (0.28)	2.13 (0.48)[a]	***P* < .001**	.112	Moderate
DJ flight, s	0.506 (0.050)		0.474 (0.057)[a]	0.474 (0.056)[a]	0.465 (0.040)[a]	***P* < .001**	.072	Moderate
DJ contact, s	0.231 (0.024)		0.247 (0.041)	0.246 (0.036)	0.238 (0.032)	.473	.041	Small
MRJ leg stiffness, kN·m^{-1}	59.60 (9.68)		57.20 (15.91)	63.21 (13.50)	59.28 (15.31)	.431	.026	Small
MRJ flight, s	0.341 (0.083)		0.183 (0.031)[a,c,d]	0.362 (0.103)	0.316 (0.093)	***P* < .001**	.466	Large
MRJ contact, s	0.174 (0.020)		0.180 (0.020)	0.175 (0.016)	0.180 (0.052)	.152	.053	Small
Cognitive performance (part II)								
Mean RT ≤ 500 ms	272 (18)	290 (26)[a]	279 (21)			***P* < .001**	.118	Moderate
Mean RT, ms	289 (33)	331 (116)	289 (28)			.093	.077	Moderate
10th percentile RT, ms	292 (387)	305 (258)	246 (166)			.867	.006	—
90th percentile RT, ms	239 (14)	256 (22)[a]	247 (15)			***P* < .001**	.156	Large
Number of false starts	3 (2)	2 (2)	3 (3)			.945	.003	—
False starts RT, ms	0.084 (0.072)	0.353 (0.969)	0.084 (0.083)			.235	.057	Small
Number of lapses	0.0 (0.0)	0.1 (0.3)	0.0 (0.0)			—	—	—
Mean lapse RT, ms	144 (253)	143 (293)	85 (135)			.891	.014	Small
Cumulative lapse RT, ms	0 (0)	30 (107)	0 (0)			.235	.058	Small
Visual analog scales (part II)								
Mental fatigue	52 (19)	65 (13)[b]	31 (16)[a]			***P* < .001**	.409	Large
Physical fatigue	56 (18)[b]	57 (17)[b]	33 (12)			***P* < .001**	.309	Large
Readiness to perform	57 (20)	28 (14)[a,b]	66 (13)			***P* < .001**	.499	Large

Abbreviations: DJ, drop jump; MRJ, multirebound jump; RSI, reactive strength index; RT, reaction time. Note: Bold type indicates significant main effects (ANOVA *P* < .05). [a]Significant difference compared with HOME (*P* < .05). [b]Significant difference compared with AWAY 1 (*P* < .001). [c]Significant difference compared with AWAY 2 (*P* < .001). [d]Significant difference compared with AWAY 3 (*P* < .001).

48 hours after arrival. While DJ contact time remained unaffected, preserving the stretch-reflex response (amortization phase of SSC),[25,26] the declined DJ flight time suggests that travel may negatively impact the concentric phase of maximal jumps. Concentric contractile force and rate of force development in the DJ are crucial for early acceleration in maximal sprints and jumps.[27] As a result, our findings unveil the immediate effects of overnight, long-haul travel on lower limb explosiveness while shedding light on the potential consequences that may deteriorate the ability to execute rugby-specific skills/movements in a fast, explosive manner.[27]

Importantly, decreased reactive jump strength (ie, reduced DJ RSI and flight time) occurred immediately after travel, and DJ flight time did not exhibit significant signs of recovery over the subsequent 2 days. Team scheduling constraints prevented us from monitoring jump performance until scores returned to baseline, which would have provided valuable insight into the recovery time required to completely restore lower limb explosiveness posttravel. Nonetheless, we add the novel observation that overnight, long-haul travel hinders leg power for *at least* 3 mornings in professional rugby union players.

In this study, MRJ leg stiffness remains unaffected by travel. However, significant reductions were observed in MRJ flight time on AWAY 1. This may be due to neurophysiological factors. Altered muscular elasticity and/or the neural control adopted during the MRJ (either voluntarily or involuntarily), resulting in suboptimal muscular activation, may have contributed to reducing MRJ flight time.[26] This aligns with observations in part II, where neurocognitive behavior diminished immediately following travel but restored capacities the morning after a night of sleep at the arrival destination. However, this highlights the need for future studies to explore the implications of travel-induced sleep disturbances on the neurophysiological deterioration of jump ability in athletes. Furthermore, future research could utilize sleep architecture (via polysomnography) and/or muscular activation assessment (via muscle imaging and surface electromyography), which could inform potential mechanisms that link athlete travel, sleep, and performance.

Overall, findings from part I suggest that overnight, long-haul travel results in partial sleep deprivation in rugby players, deteriorating their ability to perform fast SSC movements and decreasing reactive jump strength upon arrival. Despite sleep compensation on subsequent nights (hotel sleep), lower limb explosiveness remains diminished for at least 48 hours. Strategies to promote lower limb blood flow such as activation exercises, hydration, or compression garments during and/or after long-haul flights may help preserve reactive jump strength.[28–30] However, future studies need to determine their efficacy for professional rugby players to streamline protocols for preserving lower limb explosiveness for matchday success.

Cognitive Performance and Mental Fatigue

Part II presents novel empirical evidence that overnight, long-haul travel induces mental fatigue and deteriorates cognitive ability in athletes. Our data shows that travel impaired the players' capacity to respond effectively to stimuli, resulting in the slowest response latencies (mean RT ≥ 500 ms and 90th percentile RT) during TRAVEL. Increased mental fatigue provides a plausible explanation

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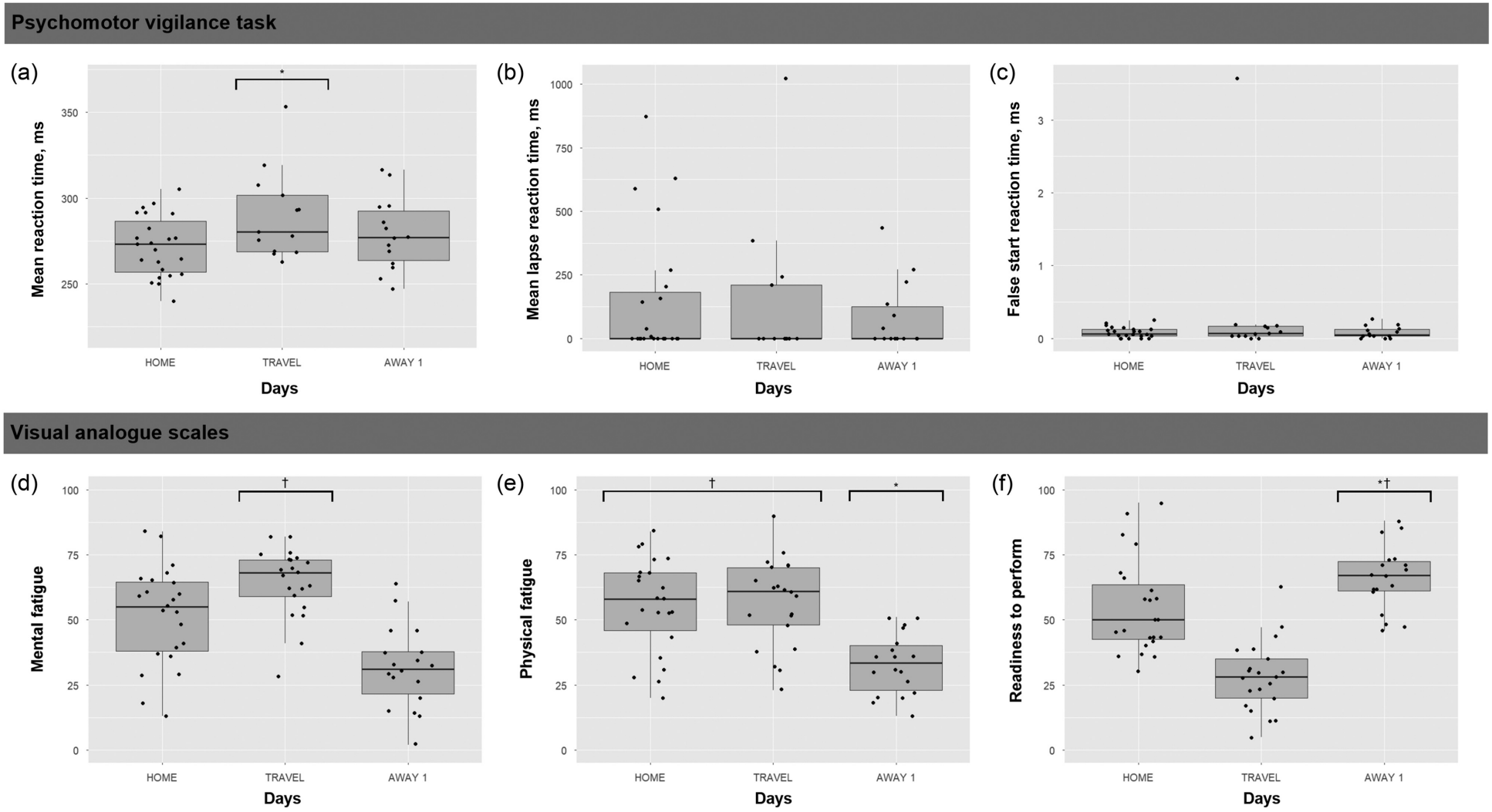

Figure 3 — Cognitive performance (a–c) and fatigue scores (d–f) before (HOME), during (TRAVEL), and after (AWAY 1) travel (part II). Outer margins of the box plots represent the interquartile range of the data, the bold horizontal lines represent the medians, the whiskers represent the minimum and maximum range 1.5 times the interquartile range of the data, and the dots represent individual data points. *Significant difference compared with HOME. †Significant difference compared with AWAY 1.

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Table 2 Sleep in Professional Rugby Union Players Before, During, and After Long-Haul Travel to 2 Match Destinations (Parts I and II)

	HOME	TRAVEL	AWAY 1	AWAY 2	AWAY 3	ANOVA *P* value	η_p^2	Effect size
Perth to Tokyo (part I), n = 24								
Total sleep time, min	421 (64)	202 (82)[a,b,c]	490 (70)	454 (40)		***P* < .001**	.753	Large
Sleep efficiency, %	79 (7)	60 (14)[a,b,c]	83 (8)	78 (5)		***P* < .001**	.493	Large
Wake after sleep onset, min	106 (34)	132 (75)[b]	97 (44)	119 (31)		**.031**	.072	Moderate
Perth to Johannesburg (part II), n = 24								
Total sleep time, min	454 (166)	163 (76)[a,b]	415 (53)			***P* < .001**	.723	Large
Sleep efficiency, %	83 (10)	77 (17)	84 (6)			.268	.062	Moderate
Wake after sleep onset, min	72 (40)	41 (39)[b]	78 (35)			**.014**	.198	Large

Note: Bold type indicates significant main effects (ANOVA $P < .05$).
[a]Significant difference compared with HOME ($P < .001$). [b]Significant difference compared with AWAY 1 ($P < .05$). [c]Significant difference compared with AWAY 2 ($P < .05$).

for the decrement in response latencies seen in the PVT, which aligns with findings from Angius et al.[10] Mental fatigue was highest during TRAVEL, elevating from HOME and reducing on AWAY 1. Findings indicate that overnight, long-haul travel can elevate mental fatigue and delay neurocognitive and psychomotor behavior, likely deteriorating the ability to sustain attention.[21]

Our data demonstrates that cognitive performance is restored by the following morning (90th percentile, mean RT ≤ 500 ms) after a full night's rest (6.9 [0.9] h sleep). This outcome suggests cognitive performance in athletes recovered after 1 night of sleep at the destination. However, the results may also reflect the time-zone change (−6 TZ) that altered participants' surrounding activities before the PVT assessment. For instance, at HOME, testing occurred at ~02:00 PM (origin time) after a full day of training (and prolonged wake). In contrast, on AWAY 1, testing was conducted in the morning (~08:00 AM local time), shortly after waking, albeit the players' circadian rhythms still being anchored to the origin time zone (~02:00 PM). The greater diurnal activity and prolonged wake before PVT assessment, in addition to the potential accumulation of mental and physical fatigue at HOME,[10] may have contributed to improving psychomotor vigilance scores on AWAY 1. Unfortunately, we did not assess cognitive performance at different timepoints, leaving it unknown whether PVT responses would remain superior to HOME if assessed in the afternoon on AWAY 1. Accordingly, more research is warranted to address the effects of travel across time zones on cognitive performance by assessing multiple days and timepoints.

Physical fatigue presented a similar pattern to mental fatigue; however, the magnitude of elevation during TRAVEL did not yield statistical significance. In this study, travel induced mental, more than physical, fatigue. Travel stressors associated with sleep deprivation, monotony in-flight (ie, boredom and prolonged seating), and worries of environment instability (unfamiliarity, safety, and food availability) at the destination likely induce mental fatigue;[11] meanwhile, airport procedures (ie, baggage handling, confined seating, and airport transfer) likely induce physical fatigue.[2]

Importantly, while cognitive and VAS scores significantly reduced upon arrival (and do not significantly differ from HOME), fatigue was not completely alleviated on AWAY 1. Notably, sleep quantity improved on arrival the first night (6.9 [0.9] h) relative to TRAVEL (2.7 [1.3] h) but remained lower than what was achieved at HOME (7.6 [2.8] h). Full restoration of response latency and VAS scores may likely have occurred with the opportunity for sufficient rest (~7–9 h).[31] While the negative consequences of playing under physical fatigue have been well-reported,[10,32] a recent study observed that mental fatigue can impair psychomotor coordination of tackle proficiency, potentially increasing the risk of injury in rugby players.[33] However, more studies should be conducted to explore the mental fatigue consequences of different travel schedules (travel duration, direction, timing, and frequency) and the psychobehavioral implications that may impact a traveling team.

In summary, part II of our study reveals that overnight, long-haul travel can induce mental fatigue and negatively influence cognitive performance (psychomotor vigilance) in traveling rugby players. If players are to undergo cognitively demanding tasks (eg, prolonged attentiveness for training meetings) upon arrival, teams may schedule these in the morning, as cognitive performance appears to be recovered the first morning after travel. However, our results show minimal residual fatigue may persist. If schedules include cognitively demanding tasks immediately upon arrival, teams may consider strategies to alleviate mental fatigue. Strategies may involve sleep extension, physiological countermeasures (caffeine and/or creatine ingestion), and/or behavioral methods (music and extrinsic motivation).[34–37] Prioritizing training variety and social connection may also help relax players and enhance mental freshness in traveling athletes while allowing physical recovery.

Limitations

Limitations of this study should be considered when interpreting results. First, the 2 travel journeys differed in direction, duration, and timing, likely inducing different degrees of jetlag and travel fatigue; therefore, the results of parts I and II cannot be interpreted collectively. Despite this limitation, the findings suggest avoiding overnight travel for competitions, especially when athletes have limited time for match preparation (~2–3 d), as is common in team sports.[1] Second, in part II, cognitive performance at HOME was collected during the afternoon (posttraining), as alluded to previously, potentially explaining the elevated baseline values due to accumulated mental and physical fatigue throughout the day. Contrastingly, scores at AWAY 1 were collected in the morning. This discrepancy is acknowledged as a limitation, and results might differ if AWAY 1 scores had been collected in the afternoon. Finally, it is worth noting that this field-based study assessed a professional team traveling for competition, where training and competition schedules took priority.

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This limited our ability to collect additional data to determine the recovery timeline of physical and cognitive performance.

Conclusion

Overnight long-haul travel induces partial sleep deprivation (~3 h in-flight, as seen in both part I and part II), likely explaining the immediate deterioration in physical and cognitive performance of professional rugby union players. While sleep was restored on the first night after travel, physical performance remained impaired for 3 mornings on arrival (+1 time zone; part I). Cognitive performance appeared to return to baseline on the first morning on arrival after travel (−6 time zones; part II), although results may vary depending on assessment timing. For teams undertaking overnight long-haul journeys for competition, it is recommended to arrive more than 3 days before matches to enable sufficient physical and cognitive recovery, thereby optimizing performance away from home.

Practical Implications

- Overnight long-haul travel significantly disrupts sleep for athletes, regardless of circadian shifts. Accordingly, teams should schedule departure and arrival times that minimize sleep disruption to preserve physical and cognitive performance for matches.
- For overnight, long-haul travel involving minimal circadian shift (similar to part I), teams should aim to arrive more than 3 days before match day to ensure sufficient recovery of physical performance (lower limb explosiveness).
- Traveling teams may schedule cognitively demanding tasks (eg, match reviews or rugby learning content) in the mornings, as cognitive performance may not be hindered after sufficient sleep at the new destination after overnight long-haul travel across multiple time zones (similar to part II). However, considering that cognitively demanding tasks can influence subsequent physical performance, allowing time for mental recovery between meetings and field sessions may be beneficial.

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References

1. Lo M, Aughey RJ, Stewart AM, Gill N, McDonald B. The road goes ever on and on-a socio-physiological analysis of travel-related issues in super rugby. *J Sports Sci*. 2021;39(3):289–295. doi:10.1080/02640414.2020.1816290
2. Janse van Rensburg DC, Jansen van Rensburg A, Fowler PM, et al. Managing travel fatigue and jet lag in athletes: a review and consensus statement. *Sports Med*. 2021;51(10):2029–2050. doi:10.1007/s40279-021-01502-0
3. Fowler P, Duffield R, Vaile J. Effects of simulated domestic and international air travel on sleep, performance, and recovery for team sports. *Scand J Med Sci Sports*. 2015;25(3):441–451. doi:10.1111/sms.12227
4. Chapman DW, Bullock N, Ross A, Rosemond D, Martin DT. Detrimental effects of west to east transmeridian flight on jump performance. *Eur J Appl Physiol*. 2012;112(5):1663–1669. doi:10.1007/s00421-011-2134-6
5. Everett KLA, Chapman DW, Mitchell JA, Ball N. Effects of westbound trans-meridian travel on countermovement jump performance in international-level rowers. *J Strength Cond Res*. 2022;36(7):1961–1965. doi:10.1519/JSC.0000000000003762
6. Fowler PM, Knez W, Crowcroft S, et al. Greater effect of east versus west travel on jet lag, sleep, and team sport performance. *Med Sci Sports Exerc*. 2017;49:2548–2561. doi:10.1249/mss.0000000000001374
7. Maciejewska-Skrendo A, Leźnicka K, Leońska-Duniec A, et al. Genetics of muscle stiffness, muscle elasticity and explosive strength. *J Hum Kinet*. 2020;74:143–159. doi:10.2478/hukin-2020-0027
8. Coghetto G, Zanrosso EM, Rabello R, da Ros JL, Rodrigues R. Association between success and unsuccess rates on technical skills and physical qualities in rugby players. *Res Q Exerc Sport*. 2023;94(1):254–262. doi:10.1080/02701367.2021.1967845
9. Merrigan JJ, Stone JD, Hornsby WG, Hagen JA. Identifying reliable and relatable force–time metrics in athletes—considerations for the isometric mid-thigh pull and countermovement jump. *Sports*. 2021;9(1):10004. doi:10.3390/sports9010004
10. Angius L, Merlini M, Hopker J, et al. Physical and mental fatigue reduce psychomotor vigilance in professional football players. *Int J Sports Physiol Perform*. 2022;17:387. doi:10.1123/ijspp.2021-0387
11. Russell S, Jenkins D, Rynne S, Halson SL, Kelly V. What is mental fatigue in elite sport? Perceptions from athletes and staff. *Eur J Sport Sci*. 2019;19(10):1367–1376. doi:10.1080/17461391.2019.1618397
12. Russell S, Jenkins DG, Halson SL, Kelly VG. Mental fatigue increases across a 16-week pre-season in elite female athletes. *J Sci Med Sport*. 2022;25(4):356–361. doi:10.1016/j.jsams.2021.12.002
13. Lim J, Dinges DF. Sleep deprivation and vigilant attention. *Ann N Y Acad Sci*. 2008;1129(1):305–322. doi:10.1196/annals.1417.002
14. Kruger A, Du Plooy K, Kruger P. Thinking differently about rugby performance: the relationship between cognitive functioning and on-field performance. *Afr J Phys Health Educ Recreat Dance*. 2018;24:250.
15. Fullagar HHK, Skorski S, Duffield R, Hammes D, Coutts AJ, Meyer T. Sleep and athletic performance: the effects of sleep loss on exercise performance, and physiological and cognitive responses to exercise. *Sports Med*. 2015;45(2):161–186. doi:10.1007/s40279-014-0260-0
16. Kölling S, Duffield R, Erlacher D, Venter R, Halson SL. Sleep-related issues for recovery and performance in athletes. *Int J Sports Physiol Perform*. 2019;14(2):144–148. doi:10.1123/ijspp.2017-0746
17. Stevens CJ, Thornton HR, Fowler PM, Esh C, Taylor L. Long-haul northeast travel disrupts sleep and induces perceived fatigue in endurance athletes. *Front Physiol*. 2018;9:1826. doi:10.3389/fphys.2018.01826
18. Tobin DP, Delahunt E. The acute effect of a plyometric stimulus on jump performance in professional rugby players. *J Strength Cond Res*. 2014;28(2):367–372. doi:10.1519/JSC.0b013e318299a214
19. Girard O, Millet GP. Physical determinants of tennis performance in competitive teenage players. *J Strength Cond Res*. 2009;23(6):1867–1872. doi:10.1519/JSC.0b013e3181b3df89
20. Lloyd RS, Oliver JL, Hughes MG, Williams CA. Reliability and validity of field-based measures of leg stiffness and reactive strength index in youths. *J Sports Sci*. 2009;27(14):1565–1573. doi:10.1080/02640410903311572
21. Basner M, Mollicone D, Dinges DF. Validity and sensitivity of a brief psychomotor vigilance test (PVT-B) to total and partial sleep deprivation. *Acta Astronautica*. 2011;69(11):949–959. doi:10.1016/j.actaastro.2011.07.015

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22. Carney CE, Buysse DJ, Ancoli-Israel S, et al. The consensus sleep diary: standardizing prospective sleep self-monitoring. *Sleep*. 2012; 35(2):287–302. doi:10.5665/sleep.1642
23. Boyne K, Sherry DD, Gallagher PR, Olsen M, Brooks LJ. Accuracy of computer algorithms and the human eye in scoring actigraphy. *Sleep Breathing*. 2013;17(1):411–417. doi:10.1007/s11325-012-0709-z
24. Cohen J. Chapter 8–F tests on means in the analysis of variance and covariance. In: Cohen J, ed. *Statistical Power Analysis for the Behavioral Sciences*. Academic Press; 1977:273–406.
25. Kenny IC, Caireallain AÓ, Comyns TM. Validation of an electronic jump mat to assess stretch-shortening cycle function. *J Strength Cond Res*. 2012;26(6):1601–1608. doi:10.1519/JSC.0b013e318234ebb8
26. Grosset J-F, Mora I, Lambertz D, Pérot C. Changes in stretch reflexes and muscle stiffness with age in prepubescent children. *J Appl Physiol*. 2007;102(6):2352–2360. doi:10.1152/japplphysiol.01045.2006
27. Cronin JB, Hansen KT. Strength and power predictors of sports speed. *J Strength Cond Res*. 2005;19(2):349–357. doi:10.1519/14323.1
28. Donahue PT, Wilson SJ, Williams CC, Hill CM, Valliant M, Garner JC. Impact of hydration status on jump performance in recreationally trained males. *Int J Exerc Sci*. 2020;13(4):826–836. PubMed ID: 32922636
29. Wannop JW, Worobets JT, Madden R, Stefanyshyn DJ. Influence of compression and stiffness apparel on vertical jump performance. *J Strength Cond Res*. 2016;30(4):1093–1101. doi:10.1519/JSC.0000000000001195
30. Li FY, Guo CG, Li HS, Xu HR, Sun P. A systematic review and net meta-analysis of the effects of different warm-up methods on the acute effects of lower limb explosive strength. *BMC Sports Sci Med Rehabil*. 2023;15(1):106. doi:10.1186/s13102-023-00703-6
31. Hirshkowitz M, Whiton K, Albert SM, et al. National Sleep Foundation's sleep time duration recommendations: methodology and results summary. *Sleep Health*. 2015;1(1):40–43. doi:10.1016/j.sleh.2014.12.010
32. Davidow D, Redman M, Lambert M, et al. The effect of physical fatigue on tackling technique in Rugby Union. *J Sci Med Sport*. 2020; 23(11):1105–1110. doi:10.1016/j.jsams.2020.04.005
33. Davidow D, Smith M, Ross T, et al. Mental fatigue impairs tackling technique in amateur rugby union players. *Int J Sports Physiol Perform*. 2023;18(9):960–967. doi:10.1123/ijspp.2023-0159
34. Proost M, Habay J, De Wachter J, et al. How to tackle mental fatigue: a systematic review of potential countermeasures and their underlying mechanisms. *Sports Med*. 2022;52(9):2129–2158. doi:10.1007/s40279-022-01678-z
35. Van Cutsem J, Roelands B, Pluym B, et al. Can creatine combat the mental fatigue-associated decrease in visuomotor skills? *Med Sci Sports Exerc*. 2020;52(1):120–130. doi:10.1249/mss.0000000000002122
36. Mah CD, Mah KE, Kezirian EJ, Dement WC. The effects of sleep extension on the athletic performance of collegiate basketball players. *Sleep*. 2011;34(7):943–950. doi:10.5665/sleep.1132
37. Åkerstedt T, Knutsson A, Westerholm P, Theorell T, Alfredsson L, Kecklund G. Mental fatigue, work and sleep. *J Psychosom Res*. 2004; 57(5):427–433. doi:10.1016/j.jpsychores.2003.12.001

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